LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Ch 8 Test (Set B): Heredity

Chapter / Topic: Heredity

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.
3. Show crosses clearly with gametes and offspring wherever asked.

SECTION A

6 marks

Choose the correct option.

Q1. Plants with round seeds (RR) are crossed with plants with wrinkled seeds (rr). The F_1 seeds are: **[1]**

(a) all wrinkled (b) all round
(c) half round and half wrinkled (d) 3 round : 1 wrinkled

Q2. How many pairs of chromosomes are there in human body cells? **[1]**

(a) 21 (b) 22
(c) 23 (d) 46

Q3. A child has the sex chromosomes XX. The child is: **[1]**

(a) a boy (b) impossible to predict
(c) either a boy or a girl (d) a girl

Q4. In the F_2 generation of Mendel's dihybrid cross, what fraction of plants show both recessive traits (wrinkled and green)? **[1]**

(a) 1/16 (b) 3/16
(c) 1/4 (d) 9/16

Q5. Mendel carried out his experiments on: **[1]**

(a) fruit flies (b) garden peas
(c) wheat (d) mice

Q6. **Assertion (A):** There is very little variation among offspring produced by asexual reproduction. **Reason (R):** In asexual reproduction, small errors in DNA copying are the only source of variation. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. Why were all the F_1 plants tall, with no medium-height plants, when Mendel crossed pure tall and pure short pea plants? **[2]**

OR

Define genotype and phenotype, with one example each.

Q8. A man with blood group A marries a woman with blood group O, and their daughter has blood group O. Is this enough to tell which trait, A or O, is dominant? Explain. **[2]**

SECTION C

6 marks

Q9. A pea plant with violet flowers (Vv) is crossed with a plant with white flowers (vv). **[3]**

(i) Write the gametes produced by each parent. **[1]**

(ii) Write the genotypes of the offspring and their ratio. **[1]**

(iii) Write the phenotypic ratio. **[1]**

Q10. How do Mendel's experiments show that (i) traits may be dominant or recessive, and (ii) traits are inherited independently? **[3]**

OR

How is the number of chromosomes kept the same from one generation to the next? What is the role of germ cells?

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) Explain Mendel's dihybrid cross between round yellow (RRYY) and wrinkled green (rryy) plants, up to the F_2 phenotypes and their ratio. **[3]**

(ii) Which new combinations appear in F_2, and what do they prove? **[2]**

OR

(a) Explain how sex is determined in (i) human beings, (ii) some reptiles, (iii) snails. [3]
(b) Is the sex of a human child decided by chance? Explain. [2]

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. Earlobes can be free or attached. Assume that free earlobes (F) are dominant over attached earlobes (f). Both of Arjun's parents have free earlobes, but Arjun has attached earlobes. **[4]**

(i) What is Arjun's genotype? **[1]**

(ii) What must the genotypes of his parents be? **[1]**

(iii) What is the probability that their next child will have attached earlobes? Show the cross. **[2]**

~ All the best! ~